# Metamorphosis as a Musical Algorithm

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## Abstract

*This paper presents a personal inquiry into processes of musical transformation open to the electro-acoustic composer of today. The case study is the writer's own compositional formulae drawing from acoustic and electro-acoustic examples. A transformative process of metamorphosis is raised as the thesis and it's various subdivisions described. The elements of electronic composition are discussed in relation to the concept of metamorphosis – by filtering techniques such as wave synthesis, additive synthesis, granular synthesis, as well as frequency modulation, in the attempt of 'differentiating' and 'integrating' sound. The ratio of the Golden Section used in formal musical structure as well as in sound formation is also discussed. The genres of Musique Concrete and SoundScape provide preliminary topics for electronic formal structure before the main antithesis of this paper is raised; the electronic interactive genre that is available to the post-modern composer. Examples from the composer's Second Symphony are used as a summary for both thesis and antithesis.*

## Key Words

Metamorphosis, Golden Mean, Interactive Electronic Music, MAX, Electronic Sound

## Introduction

By inquiry into the compositional processes open to the electro-acoustic composer of today, I will attempt to provide a critique of my own adopted methods of metamorphosis and display its relevance to musical investigation. To elucidate the meaning of this word, I will propose an aesthetic that preoccupies my mind; As all life is concerned with character growth and development, I like to represent this experience as musical material distorts and reforms through metamorphosis. As with life, a musical organism must, to my mind, be in constant evolution of being.

Jointly, I will discuss the marriage of acoustic and electro-acoustic fields, which has reached a new significance for today's tone poet.

## 1. A Preliminary Electronic Technique

In order to facilitate discussion into the essence of an electronic sound technique, a preliminary inquiry into the nature of its constituent ingredients needs to be conducted. The fundamental building block for sound is waveform vibration. Analysis of this basic ingredient will pave the way to electro-acoustic techniques.

Via processes of 'differentiating' and 'integrating' sound by filtering techniques, a capacity for producing a wide range of timbral sound evolution can take place. These two processes become the new tools for compositional structure. Wave synthesis, additive synthesis, granular synthesis as well as frequency modulation are all techniques that manipulate the frequency, amplitude and/or velocity of a soundwave. The process of reducing or adding harmonics to a tone is the preliminary idea behind my attempts to create a timbral metamorphosis. The range of wave manipulation from sinusoidal sound (sine wave), square waves, sawtooth waves through to white noise provides the building blocks for the

electro-acoustic musician. It seems in this sphere of electronic music that there is a return to nature and to

sound in its unmediated or primordial state. The essence of sound becomes the thesis. Rather than predetermined hierarchies of scales and harmonies, the composer discovered the nature of sound- in its pure state.

I wish to include comment on the ratio of the Golden Mean, since it is not only widely used in electronic and acoustic formats but is also a key component in my methods of composing music. Besides its use in areas of macrostructure, microstructure and cellular/motific structure, the Golden Mean ratio can also control the harmonic content of tones. John Chowning in his piece Stria written in 1977, used the Golden Section as a means of relating partials within sounds and from this time, others such as Debussy, Bartok, Stockhousen to name a few have also applied the device. Certainly the proportion can be used other than in terms of formal structure- in the manufacture of sound itself, thus potentially creating a virtual space whereby all aspects of a musical work are dictated by this proportion.

**Golden Section as MAX patch.**

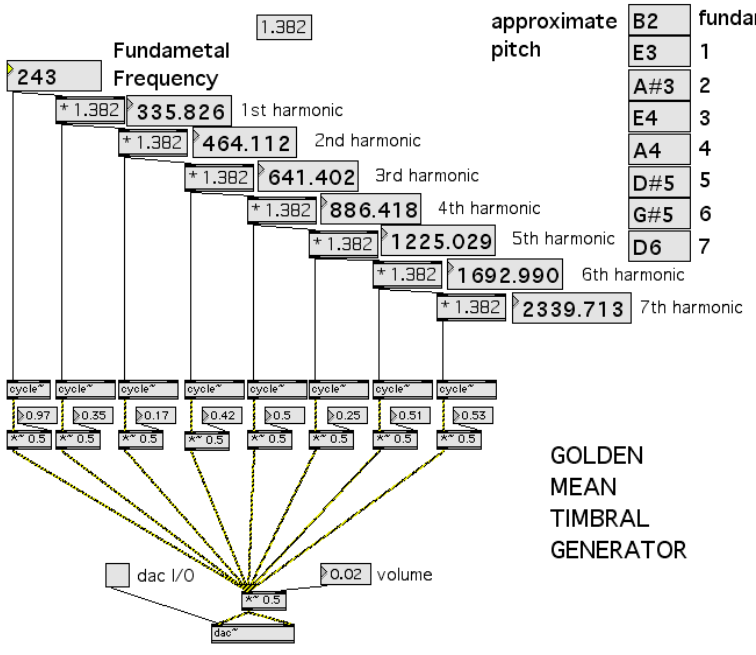



Figure 1. Max patch showing formation of tone based upon the Golden Mean.

## 2. Musique Concrete and the Post-modern Soundscape

A parent form of my metamorphic process is the sound art developments of the 1940's, that of Musique Concrete. In an attempt to contextualise my ideas of Metamorphosis, an account of these early methods as well as modern applications of Musique Concrete should be addressed.

Following a process of collecting 'sound objects' (as in the visual art image), the early attempts at Musique Concrete combined sampled resources to produce a collage-like structure with the aim of creating a pastiche of everyday life. After manipulating these samples through processes of filtering, distortion and modification, the result would often be unrecognisable. By the very nature of these samples, there is often evident an adopted symbolic representation of the original source, even when distorted from the original context. Certainly if common sound samples or other musical structures have been commonly ascertained to a culture, style or a work, the use of these components draws the ear to the cultural framework that is implied. Naturally the link between affective stimuli and cultural connotations needs to be taken into account if an intended meaning or affect is to be truly conveyed.

Of course hybrid forms employing electronically produced sounds and natural sounds are explored today. The interaction of the Musique Concrete principal and pure electronic sound sources is a valid form of compositional hybridity open to the post-modern composer. The term for this hybrid form is the SoundScape, and as the name implies, a virtual space is constructed by a localisation of sound.

## 3. Metamorphic Processes

Before describing my metamorphic processes and their various applications to electronic and/or acoustic genres, it is pertinent to provide an introduction exploring the nature of transformation. An elementary definition of the word metamorphosis is a logical beginning. Hence the following two descriptions; firstly, a complete change of form or substance occurs and, secondly, a structural principle which allows a small cell to expand into an organic, self-perpetuating whole. Two approaches are immediately apparent through these two definitions. Firstly, something already formed gradually changes into another form. Secondly, a sense of growth is witnessed by the duplication of cells (cell division) from a single cellular source.

Two paradigms of contrasting origins have influenced my concept. The first belongs to the natural organic growth and mutation of a cell from the natural sciences, and the second derives from the techniques of development within the Sonata Form of western music. In order to gain full understanding of my musical research, preliminary discussions concerning this second paradigm need to be undertaken before examples of my processes exploring both can be given.

To validate the inclusion of such a topic as Sonata Form in this paper, I will provide a brief account of the validity of historical inquiry, regardless of the discipline. It is of paramount importance to criticise any new philosophical law or artistic endeavour in light of its historical antecedents, since all generations of subsequent achievement are direct or indirect results of past developments. Of course the genesis of Sonata Form really grew from the 18th century aesthetic of rational thought and the logical exploration of an idea. The proposition of hypothesis and the subsequent examination towards supporting this hypothesis was central to the idea of debate so influential to the 18th century psyche. Naturally there were musical works bearing the title of the Sonata before this time and these earlier formal developments naturally influenced subsequent formats, but the evolution of the sonata form technique can be seen as owing its genesis to 18th century science and philosophy.

The central idea influencing this philosophy was the thesis and antithesis- the proposition of an idea that is compared and contrasted to another idea whereby an ordered argument is set up before reaching a conclusion. The philosophy can relate directly to the structure of sonata form in the following way; a statement of the thesis in the tonic key is followed by the contrasting antithesis (second subject) in the dominant key. The process of debate though justification and comparison can be seen in the subsequent development section whereby fragments of the thesis and antithesis are extended through a variety of interpretations. The recapitulation, thus, can be seen as a restatement of the original conceit and the argument explained and/or opinions validated- in musical terms, the subject and second subject are both stated in the tonic key. In all, the third section can be construed as a summary of the progression. Though perhaps simplistic to the modern psyche, one can witness the sonata form's relevance to the development or growth of a musical idea.

I return to my metamorphic principle. Preliminary musical developments in metamorphosis first began for me in an acoustical context. Starting with a musical cell, motif, or texture, I formulated procedures through which the musical structure witnesses growth resulting in a complete change in musical information and thus giving rise to a completely new theme, motif or texture. The process is gradual and occurs in a step by step fashion whereby each stage of development moves logically on to the next. Below is a simplistic numeral representation showing five stages of this process;

A ABA AB BAB B.

With B representing completely new material, the whole progression reflects the gradual incorporation of B, beginning in the second stage, the equal interaction of the two forms in the third (AB), the prominence of B as A loses its character in the fourth (BAB) and the final establishment of the new material in the fifth (B).

Taking this concept as a departure point for musical structure, I have formulated four methods of transforming musical material. A further three methods specifically applicable to electronic metamorphosis will be discussed later. The first is

concerned with the simple metamorphosis of a theme/motif or acoustic texture. Utilising the intervallic and rhythmic characteristics of musical stimuli, the material gradually adopts new characteristics, almost imperceptibly, until the new foundations rise in prominence and dictate the new musical theme/motif or texture. Below this is exemplified.

## Musical Example 1

Extract from Morgan s  Symphony No 1 — Genesis
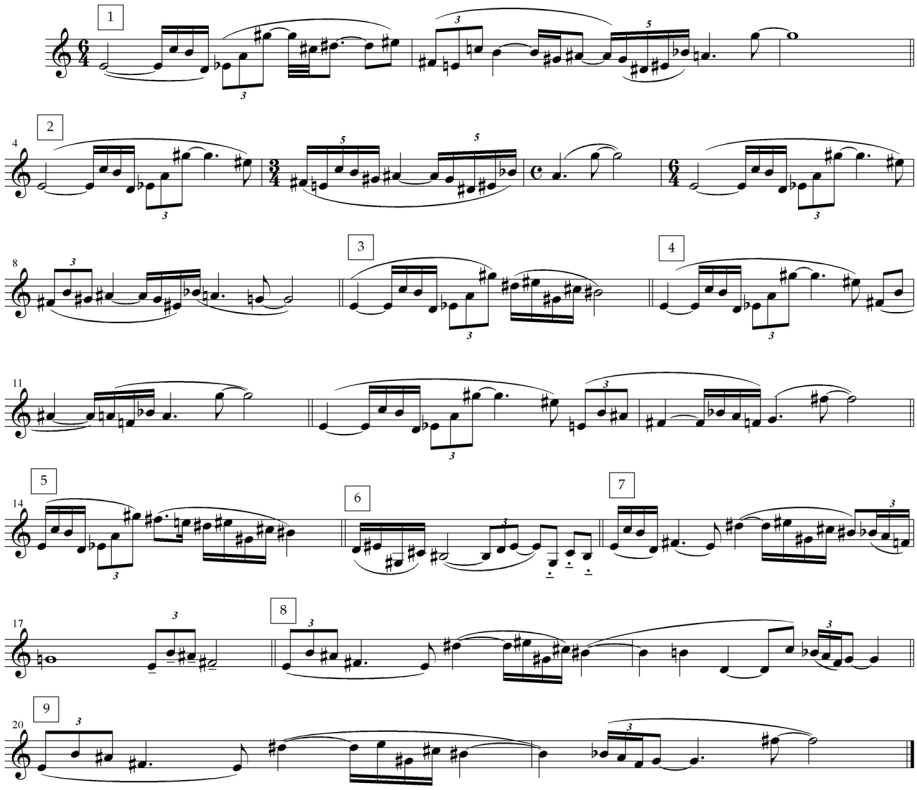


Figure 2.  Metamorphosis of a Theme.

Stages of Metamorphosis for Figure 2;

1. Theme
2. Theme is simplified.
3. Characteristic cell is isolated with a cell from the new theme.
4. The theme starts to be modified.
5. Material is reduced with the inclusion of the new cell.
6. The new cell dominates the musical line.
7. 2nd cell of new theme is introduced. Traces of original material are few.
8. First section of new theme is linked to second via a bridge from the old material.
9. Presentation of new theme.

The second method of transformation involves corresponding two thematic germs to a musical evolution, whereby the two gradually adopt similar characteristics and finally merge to create one final whole containing influences from both the originals. Furthermore, these two musical thumbprints can be stated separately and then developed side by side as they influence each other. The resolution of this method will logically be the fusion of both thematic germs to create a single new theme containing related parts from both originals. The process can be seen as one of unification- the interaction of the two, how they influence one another and finally end in a union. Below is an example from my own work whereby this system is in place.

**Musical Example 2**

Metamorphic union of two musical lines.
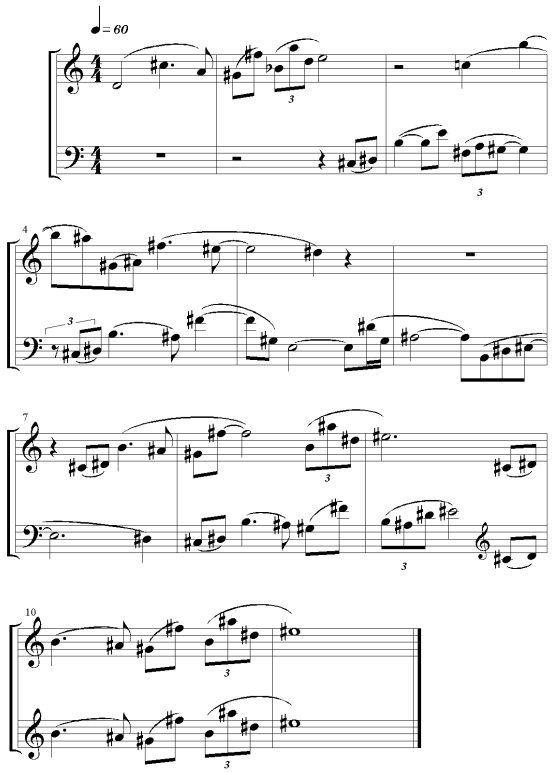


Figure 3. Excerpt from Morgan s first ballet Orlando .

The third method could be labelled Micro-metamorphosis, whereby the actual process of musical growth begins from pointillistic random stimuli. Key cellular components are established out of this milieu that gradually rises in prominence. The music then leads to the formation of a musical cell, motif or a much longer musical line. The following musical analysis is derived from my second Symphony subtitled 'Dialogues for Chamber Orchestra and MAX', whereby this technique is utilised using a live orchestra and a computer using the MAX environment. The start of the work is unformed and thus the first section is concerned with the growth of a musical cell or group of cells arising from out of a random musical chaos. MAX's random capabilities are utilised here until an organic conception is established. As a work in progress, the symphony aims to explore the culmination of individual voices by the interaction of acoustic orchestra and computer manipulated sound samples derivative of the instruments themselves.

**Musical Example 3**

Stage 2 of Micro-metamorphosis

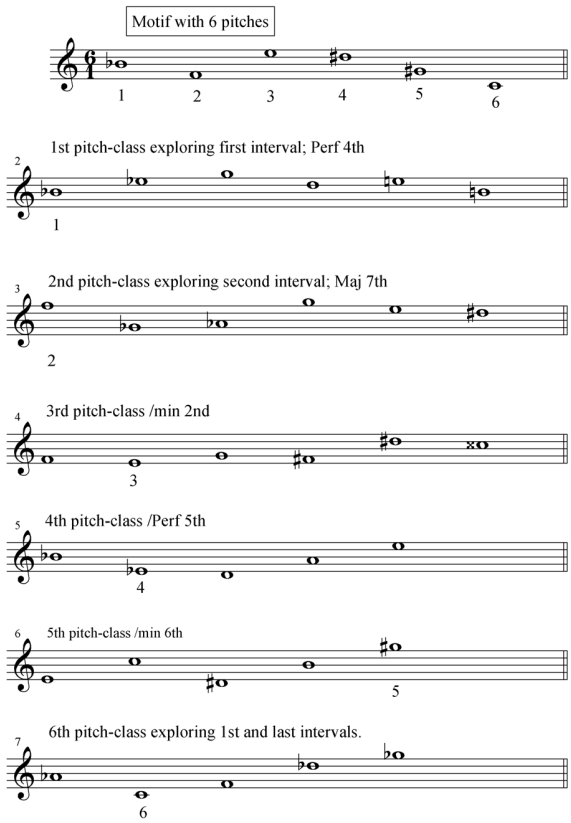



Figure 4. Excerpt from Morgan's Second Symphony- stage 2 of micro-metamorphosis.

Stage 1 of the process is concerned with the random incorporation of all 12 pitch-classes and all intervallic permutations. Stage 2 starts the filtering process whereby the six pitches and five intervals of the theme dominate. Stage 2 witnesses the reduction of pitch-classes to the six of the theme. As these occur more frequently, other pitch-classes become infrequent. Stage 3 is the presentation of the theme's intervallic ordering. Stage 4 applies rhythmic details to the pitches, leading to the final product- before this stage, no rhythmic characteristics are established.

The fourth method is concerned with the transformation of a texture. The permutations of this method are obvious. Further inquiry into textural metamorphosis will take place when electronic treatments are discussed.
To summarise, it is important to contextualise these procedures in light of musical form. Naturally these methods or techniques can be applied to a moment in the musical work, a section or they can be used as a means to construct a complete macrostructure.

Musical language and gesture become of paramount concern in this unstable and evolutionary progression. The level of harmonic dissonance can be used as a tool for creating tension as the thematic development progresses. The result, I hope, gives one the sensation of witnessing a growing, writing, thirsting musical organism constantly flowing through time and its enveloping energies.

## 4. Electronic Metamorphosis

So far, issues of musical transformation raised have been confined to the manipulation of pitch sets and the tonalities they imply. The whole realm concerning the manipulation of soundwaves lends itself to further discussions in relation to the application of the above principles.

The parameters in which the electro-acoustic musician can manipulate sound stimuli are summarised in the ideas of Stan Tempelaars. Tempelaars differentiates between 'global' modulatory sound (whereby the entire sound is affected) and micro-modulation (which causes change instant to instant). Furthermore, the developments of the 1990's gave rise to the term 'frequency modulation'. The following quotation from Electric Sound by Chadabe is a valuable concept for the manipulation of a waveform frequency via 'peak deviation';

In *frequency modulation*, the instantaneous frequency of one waveform, called the *carrier*, is varied by another waveform, called the *modulator*. The extent to which the carrier frequency varies, called *peak deviation*, is determined by the amplitude of the modulator. The rate at which the carrier frequency varies is determined by the frequency of the modulator.

When carrier and modulator are at audio frequencies, audible extra partials, called *sidebands*, appear in the carrier spectrum. These sidebands are located symmetrically above or below the carrier at intervals equal to the modulator frequency….The number of sidebands with significant amplitude is determined by the *modulation index*, which is the ratio between the peak deviation and the modulator frequency (index = peak deviation/modulatory frequency) (Chadabe 1997 p251).

I return to my ideas of metamorphosis. Trevor Wisthart used processes of sampling birdsong, machine sounds and other various sounds taken from the real world. Symbolic representation was then associated to the sounds just as in any other Concrete work. More interesting in this example is the juxtaposition of samples and gradual transformations between them; "A flock of birds, for example, transforms into a rustling of a book's pages, then into the sound of closing a book, then into a slammed door, into a creaking door…"(Chadabe 1997 p137). Here is the epitome of the electronic transformative process.

The use of vibrato to imply timbral change is another method. By applying a vibrato of increasing range and depth to a tone, the illusion of timbral fluctuation is evident, giving the implication of an evolving spectrum of timbre. The application of vibrato is a unique way of creating a depth of sound and by controlling the speed of the oscillation and amplitude manipulation, the technique can be utilised as another approach to metamorphosis.

That master of sound-art, Jonathan Harvey summarises these ideas of timbral transformation used as a structural device; "The timbral combinations were very interesting because in each melody the timbre changed- the shakuhachi turned into an oboe, for instance- so mixing melodies was also mixing timbres"(Chadabe 1997 p130). The implications of this statement are obvious – the range of timbral experimentation is infinite, especially considering the developments in hardware and software since 1990, the time of Harvey's observation.

To elucidate the effect of the above upon my own creative work, I again return to the word

metamorphosis. I would like to offer further definitions to the word, summarising the individual processes I utilise in the act of musical composition.

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**Metamorphosis of Tone**. Manipulating a clearly defined pitch. The transformation to a more complex array of partials can finally result in noise. Or another example, a whole combination of tones create a texture that is slowly raising (or lowering) in intonation

## Metamorphosis of Timbre.

Transformation of a specific timbral pitch to another timbre occurs by manipulating the harmonic content of the pitch.

## Textural Metamorphosis.

Single tones grow into chords. This process was used in much of my earlier acoustic works. The expansion from a unison pitch to a thick instrumental texture explores various shades of density that shift and fluctuate. A further density transformation can be found in the shifting of a soundscape texture where several timbral voices all shift to one timbre. Or a polyphonic texture utilising only one interval, say a major second, widens until all voices reach another interval, say an augmented 4th. Certainly Per Norgard's Infinity Hierarchies relate to this approach, or the textural growth
that is evident in Ligeti's works. This process also lends itself to a wide range of variation – one being the reduction of a texture incorporating all the 12 semitones to five defined pitch classes. This particular example also shows the golden ratio in operation- 12/5 = 0.618/0.382.

**Musical Example No 4**
Textural Metamorphosis
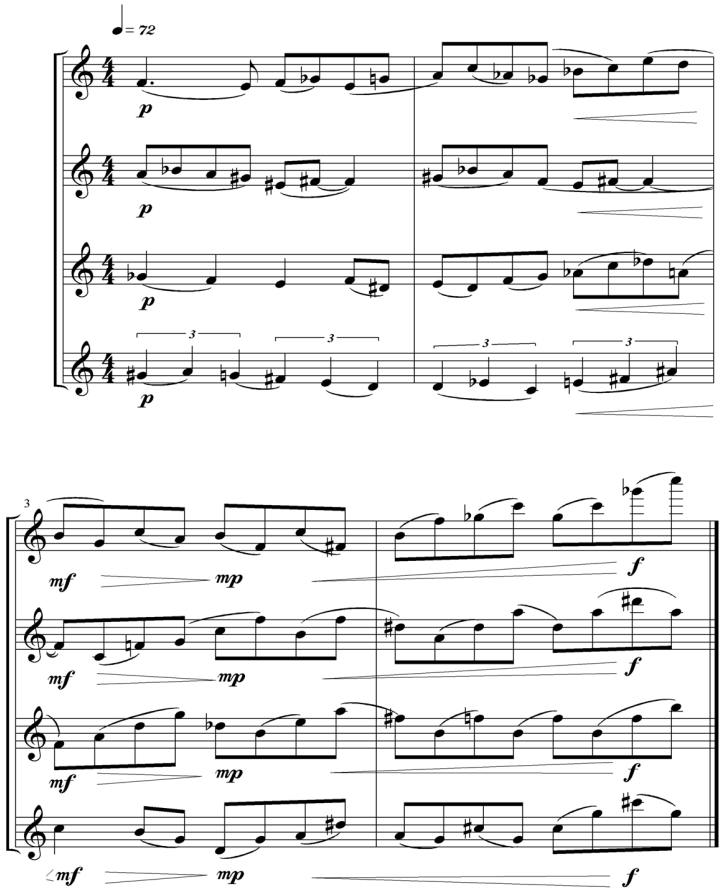



Figure 5. The expansion of a texture over four bars from a minor 2rd to an Augmented 4th.

**Rhythmic Metamorphosis**.This last form leaves the treatment of pitch, timbre and harmony. All permutations of rhythmic transformation are available. Techniques I have adopted are various rhythmic accelerations and retardations and layered rhythmic metamorphosis, whereby one layer of musical fabric starts to travel at a different speed and then all other layers speed up from that departure point. Further more, this process can be used to set up a new tempo with a new rhythmic pulse and beat subdivision. The new tempo in relation to the old can be a predetermined ratio; for example a two to three ratio, or the golden mean ratio. In musical terms, an example would be the shift of pulse from a quaver to a quaver triplet.

All of the above processes exploring the theory of metamorphosis can be utilised as passing compositional gestures and are seen in many existing works. My approach is to use them as a means of travelling through the musical structure - from one section to the next. I am also preoccupied with the rate of change; emphasising the journey before arriving at a new destination gives the new section more meaning and definition. In an attempt to summarise all my categories of the word metamorphosis, I propose the following hypothesis;

> While a musical section is established, the introduction of a new element takes root (causing a sense of conflict) and is perpetuated as it grows in importance before dominating and finally causing all material to follow and adapt until the new section based on the new element is secured.

## 5. A last note on Interactive Music- MAX and the Acoustic Orchestra

A fashionable artistic association of today is the performance-orientated collaboration of the musician with a computer- one that certainly lends itself successfully to my adopted techniques. The combination of sampled art, electronic synthesis, acoustic performance and live interaction produces a completely new aesthetic. Certainly with the rise of the software application MAX and its MSP enhancement, infinite
compositional and multi-media possibilities now exist in an interactive context. It is also important to acknowledge John Chowning and Todd Winkler and their contribution to computer interactive music. To conclude, I will contextualise the issues raised in this paper by relating them to this interactive environment.

My proposed composition Symphony No 2 'Dialogues for Chamber Orchestra and MAX' will explore the duality of highly detailed construction (various processes of metamorphosis permeated by the Golden ratio) and sections of *Maxian* Improvisation. As the ensemble is pitted against MAX, samples of the section will be utilised in the form of a computer generated improvisation using MAX's random related objects (guided by the operator who is following certain text instructions written into the patch).

Thus, as the thematic material progresses, the computer manipulates this data to produce an electronically produced synthetic response, until the compositional algorithms take over once more. As the computer cognises the order, timing and timbre of the pitches played by the instrumentalists, the principle of metamorphosis follows two independent streams of musical activity guided by the one set of musical material. As stated earlier, the cognitive modelling of the musical stimuli will be confined to certain parameters dictated by the operator and the messages incorporated into the patch -the result is somewhat like the cadenza within a concerto. Episodic in form, the work will move from sections of organised developments to those of guided improvisatory exploration.

One can see the relevance of interactive music to my ideas of Sonata Form and the 18th century aesthetic. The interaction of two sentient beings that share and develop thoughts before leading to conclusive hypotheses is the essence behind both paradigms. In the context of my second symphony, the aesthetic produced should be one whereby an isolated individual (MAX) explores a relationship with a group of individuals (the instrumental ensemble) that are all connected in some sense of censorship. How they influence one another as they converge and diverge is the thesis of this compositional style.

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